Numerical simulation of coupled cell motion and nutrient transport in NASA's rotating bioreactor

High-level overview: Chao & Das (2015), Chem. Eng. J.

Momentum balance

$$\rho\dot{\mathbf{u}} + \rho(\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla p - \nabla\cdot[\rho c_s(1-c_s)\mathbf{u}_{slip}\mathbf{u}_{slip}] + \nabla\cdot[\mu(\nabla\mathbf{u}+\nabla\mathbf{u}^T)] + \rho\mathbf{g}$$

Physical meaning

- Mixture momentum balance: fluid and suspended cells move as one continuum
- Variable-density, variable-viscosity Navier-Stokes, plus a stress from slip between the two phases
- Buoyancy enters through $\rho\mathbf{g}$

Terms

ρ, $\mathbf{u}$, p, μ, $\mathbf{g}$: mixture density, velocity, pressure, viscosity, gravity

c_s: local mass fraction of solid

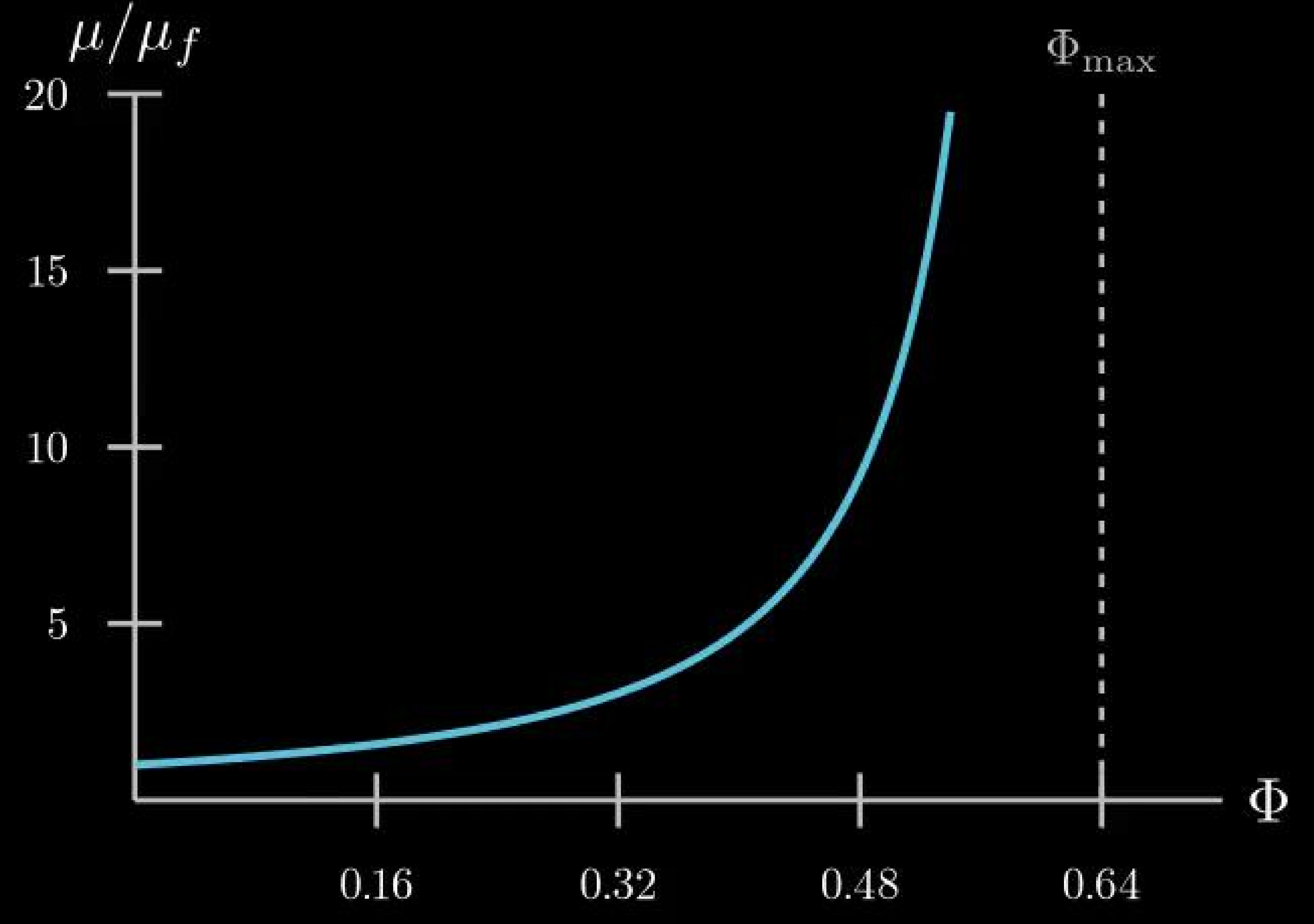
$\mathbf{u}_{slip}$: relative velocity between solid and fluid

Mixture density and viscosity

$$\rho = (1-\Phi)\rho_f^\circ + \Phi\rho_s^\circ \qquad \mu = \mu_f\left(1 - \frac{\Phi}{\Phi_{\max}}\right)^{-2.5\Phi_{\max}}$$

Physical meaning

- Density: a volume-weighted average of the pure fluid and solid densities
- Viscosity: the suspension thickens sharply as Φ approaches $\Phi_{\max}$ (jamming)



Terms

Φ: particle volume fraction (0 to $\Phi_{\max}$)

ρ_f°, ρ_s°: pure fluid and solid phase densities

μ_f: pure fluid viscosity

$\Phi_{\max}$: maximum packing fraction

Particle transport (master equation)

$$\dot{\Phi} + \mathbf{u} \cdot \nabla\Phi = -\frac{\rho}{\rho_s^\circ \rho_f^\circ} \nabla \cdot \mathbf{J}_s$$

Physical meaning

- Φ is carried along by the bulk flow $\mathbf{u}$ (advection, left-hand side)
- Plus a correction from any relative motion between the phases, via $\nabla \cdot \mathbf{J}_s$
- This is the equation that actually says where the cells end up

Terms

$\mathbf{J}_s$: particle migration flux (defined next slide)

Flux closure: shear-induced migration

$$\mathbf{J}_s/\rho_s^\circ = -\Big[\, 0.41a^2\Phi\nabla(\dot\gamma\Phi) + 0.62a^2\Phi^2\dot\gamma\nabla(\ln\mu) \,\Big]$$

$$+\,\Phi\,\frac{2\mu_f a^2(1-\Phi_{avg})(\rho_s-\rho_f)}{9\mu^2}\,\mathbf{g}$$

Physical meaning

Shear term: cells drift away from high-shear regions, toward calmer flow

Viscosity term: cells drift away from thick (viscous) regions, toward thinner ones

Settling term: gravity still pulls denser cells down, just slowed by the crowd around them

Terms

a: cell radius

$\dot\gamma$: shear rate (solved as its own field; see Implementation)

Φ_{avg}: domain-average particle volume fraction

Nutrient transport

$$\dot{C} + \mathbf{u} \cdot \nabla C = D_f \nabla^2 C + r_c$$

Physical meaning

- Standard advection, diffusion, and reaction for dissolved nutrient concentration
- r_c is a sink: nutrient is consumed by the cells

Terms

C: nutrient concentration

D_f: nutrient diffusivity

r_c: consumption rate

Growth kinetics

$$\dot{d} = k_c\, C\, d_0\, e^{k_e t}, \qquad \Phi = \frac{\pi}{6}\, d\, a^3$$

Physical meaning

- Cell number density d grows exponentially, nutrient-limited through C
- Converts back to a volume fraction Φ through the (fixed) cell radius a

Terms

d: cell number density

k_c: growth-rate constant

d_0: initial cell density

k_e: growth-rate exponent

Every equation that defines the model

Momentum: $\rho\dot{\mathbf{u}} + \rho(\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla p - \nabla\cdot[\rho c_s(1-c_s)\mathbf{u}_{slip}\mathbf{u}_{slip}] + \nabla\cdot[\mu(\nabla\mathbf{u} + \nabla\mathbf{u}^T)] + \rho\mathbf{g}$

Mixture density / viscosity: $\rho = (1-\Phi)\rho_f^\circ + \Phi\rho_s^\circ \quad \mu = \mu_f(1-\Phi/\Phi_{\max})^{-2.5\Phi_{\max}}$

Φ-transport (master eq.): $\dot{\Phi} + \mathbf{u}\cdot\nabla\Phi = -\frac{\rho}{\rho_s^\circ\rho_f^\circ}\nabla\cdot\mathbf{J}_s$

Flux closure: $\mathbf{J}_s/\rho_s = -[0.41a^2\Phi\nabla(\dot{\gamma}\Phi) + 0.62a^2\Phi^2\dot{\gamma}\nabla(\ln\mu)] + f_h\mathbf{u}_{st}\Phi$

Shear rate: $\dot{\gamma} = [\frac{1}{2}(\dot{\boldsymbol{\gamma}}:\dot{\boldsymbol{\gamma}})]^{1/2}$

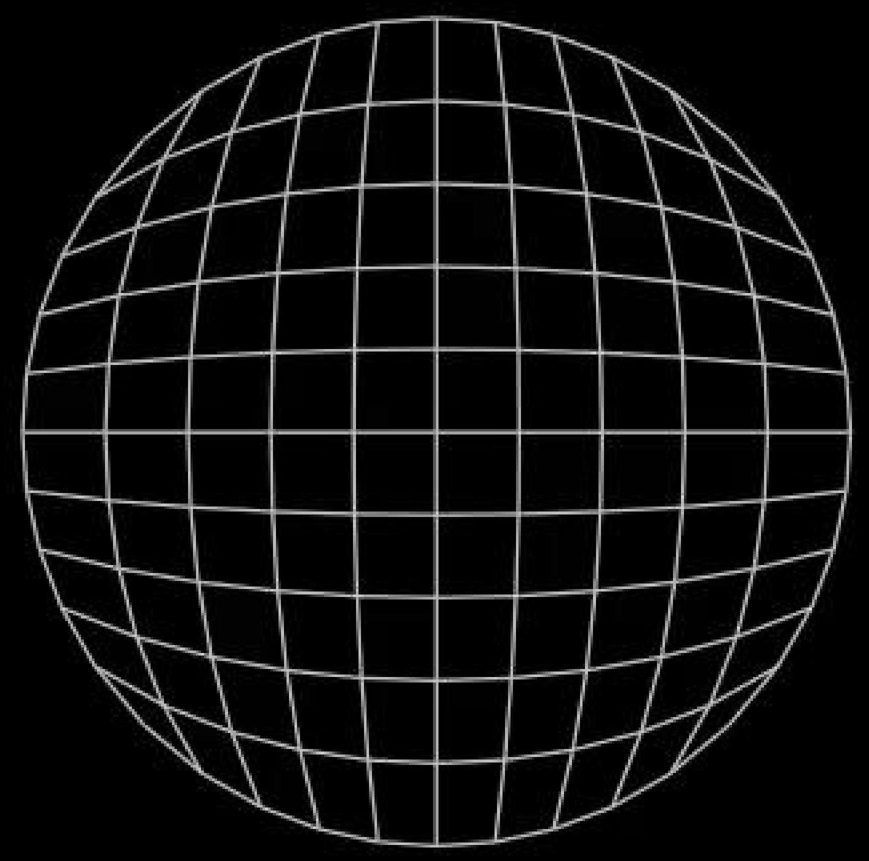
Nutrient: $\dot{C} + \mathbf{u}\cdot\nabla C = D_f\nabla^2 C + r_c$

Growth: $\dot{d} = k_c\, C\, d_0\, e^{k_e t} \qquad \Phi = \frac{\pi}{6} d\, a^3$

Monolithic finite-element solver

Solved as one weak (finite-element) system, all five fields together

Julia · Gridap.jl



$\mathbf{u}$: fluid velocity

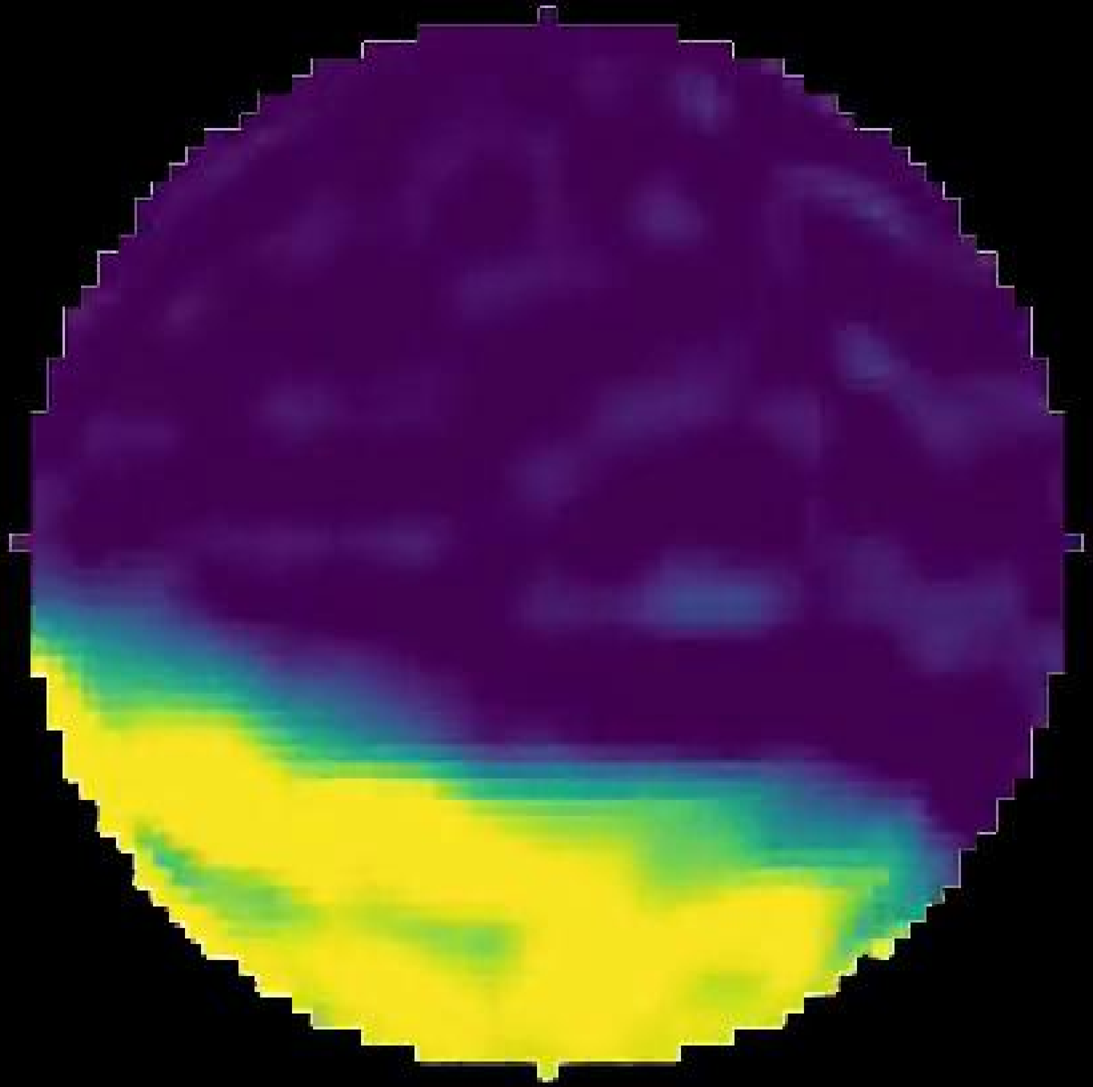
p: pressure

Φ: particle volume fraction

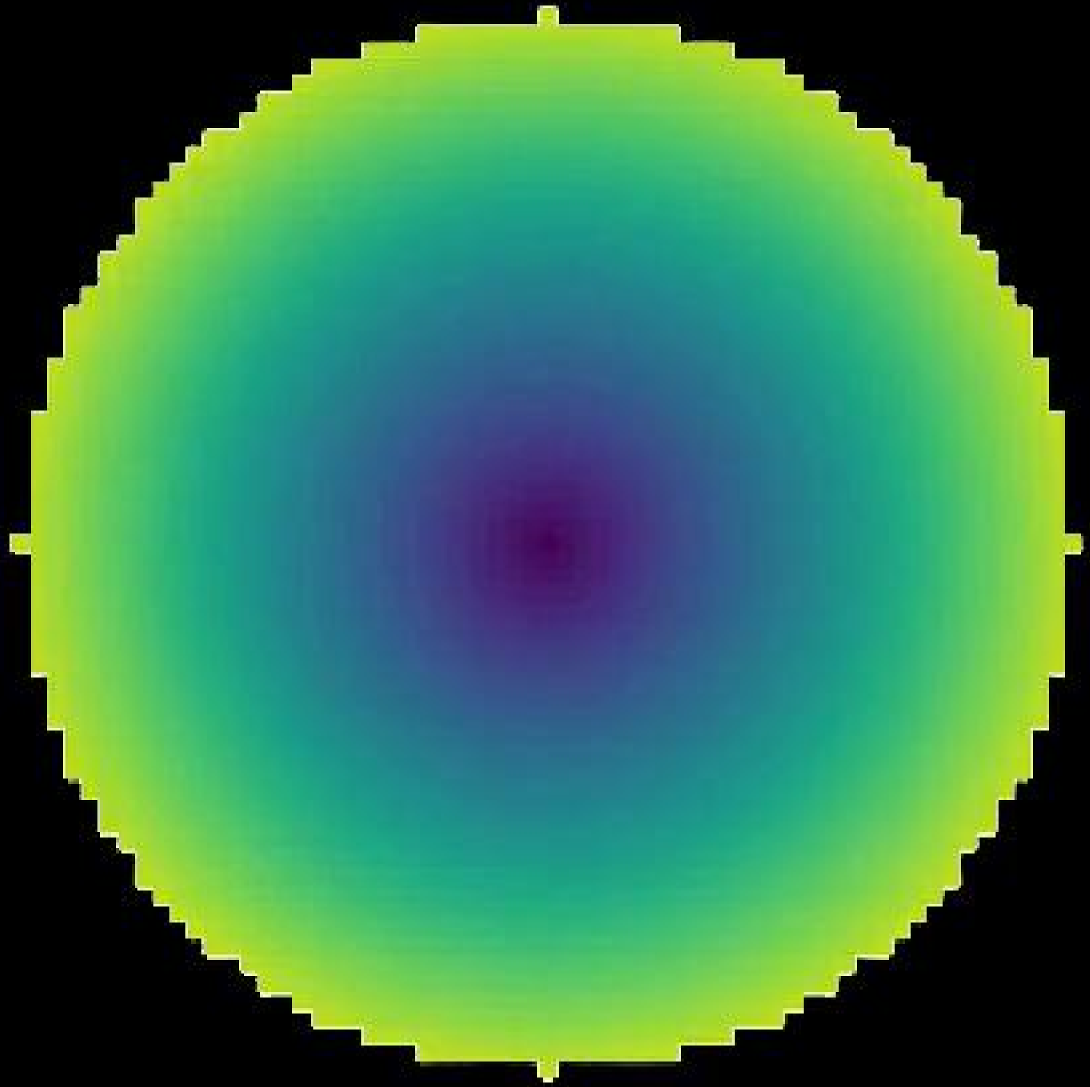
C: nutrient concentration

$\dot{\gamma}$: shear rate, solved as its own field

github.com/elvis-aguero/sbm-bioreactor



$\Phi(t)$: particle concentration



$|u|(t)$: flow speed

Simulations from the paper

Results

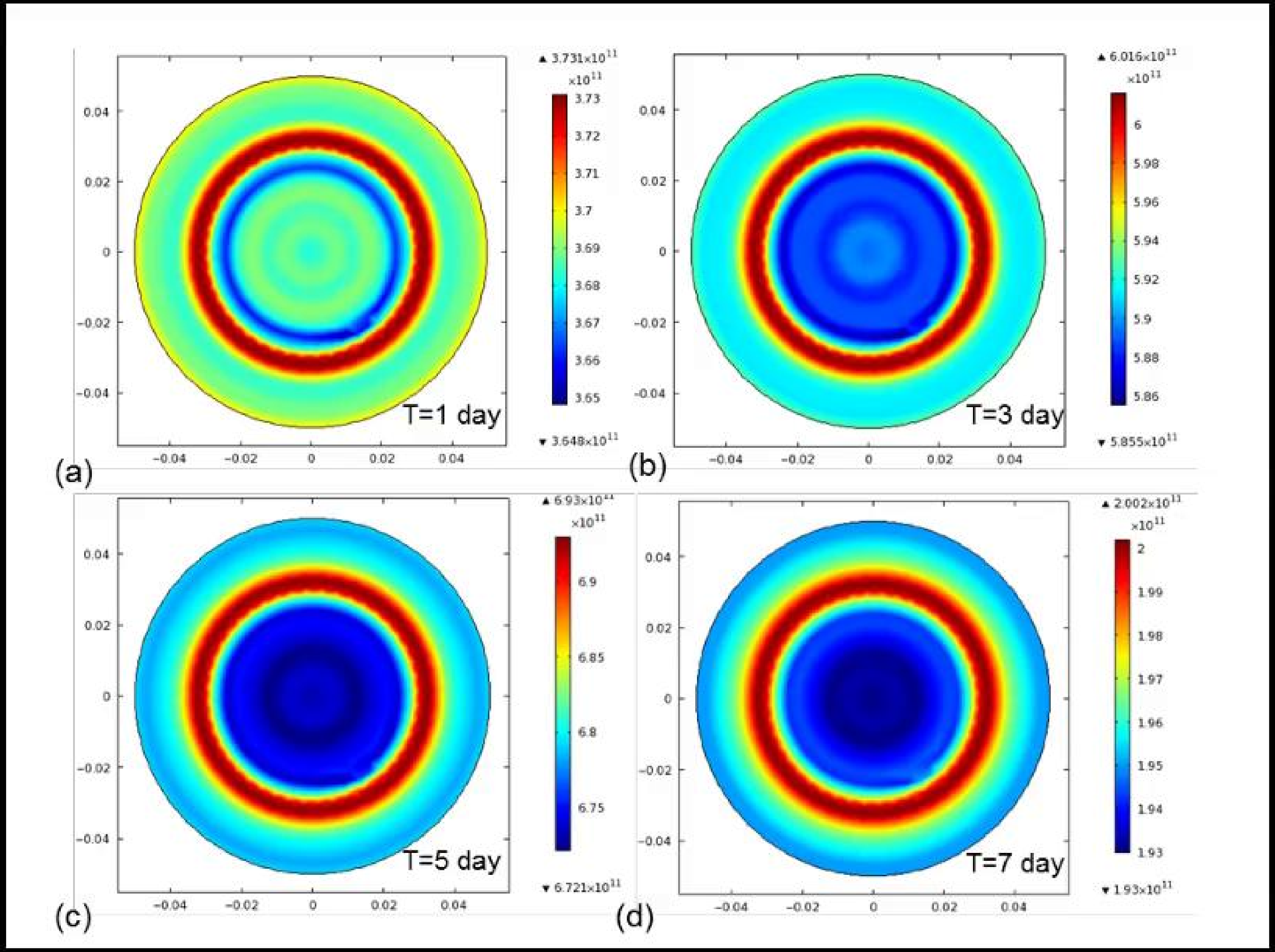


Fig. 7, Chao & Das (2015): simulated cell density at t = 1, 3, 5, 7 days

- Density concentrates in a thin, bright ring at roughly two-thirds of the disk's radius, in all four snapshots
- Just inside that ring, density is depleted: a visibly darker band sits between the ring and the center
- Shear-induced migration pushes cells out of the middle and out from the rim, and they pile up at that ring in between

Cell density peaks, then collapses

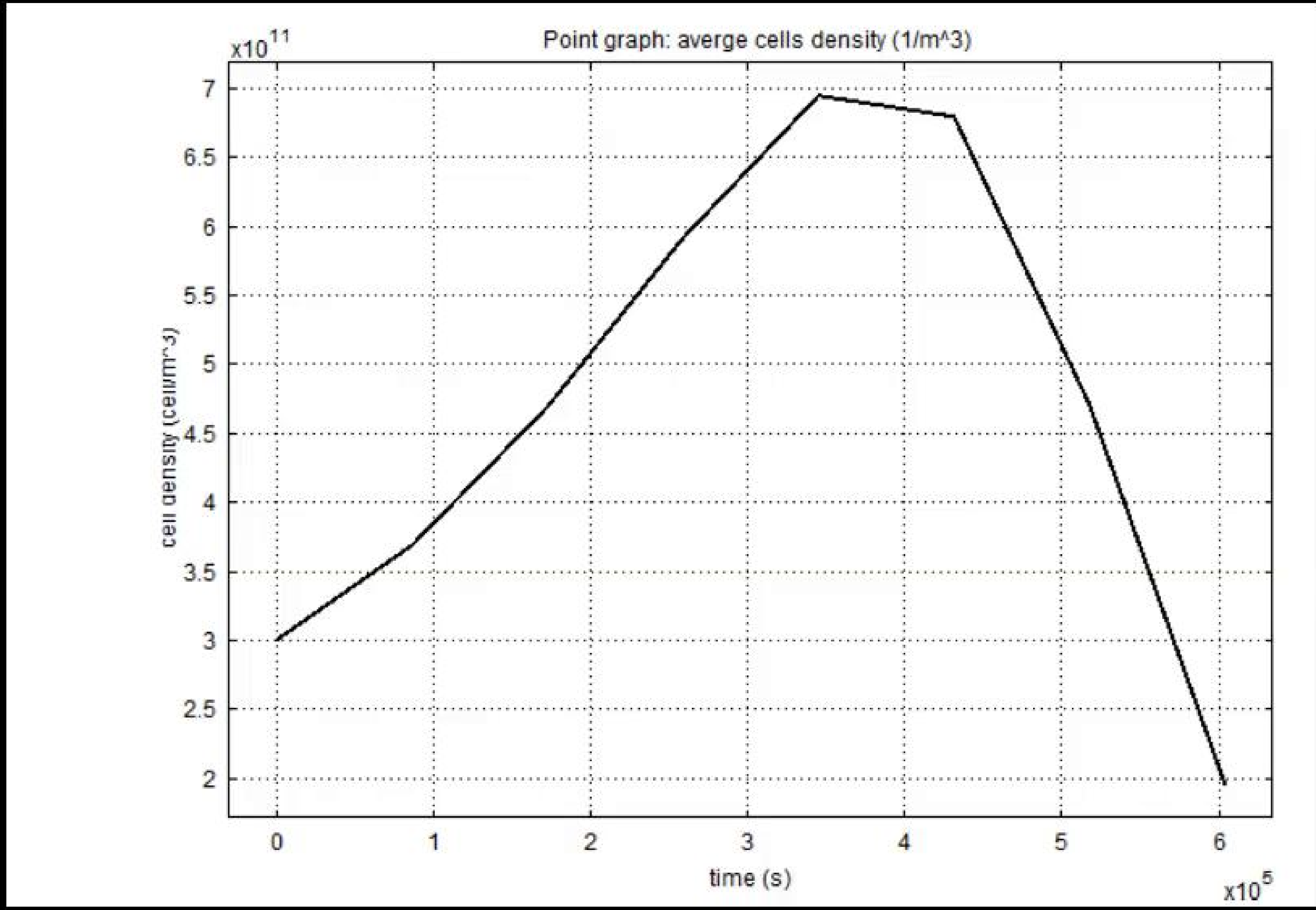


Fig. 10, Chao & Das (2015): average cell density over the same run

- Density roughly doubles over the first 4 days (about $3 \times 10^{11} \rightarrow 7 \times 10^{11}$ cells/m^3)
- It plateaus around day 4-5, matching Fig. 7's own day-5 peak
- Then it collapses by nearly two-thirds by day 7, the same crash Fig. 7's color scale hinted at, now unmistakable